



UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 17307

LM SUPERCONDUCTIVITY

SUMMER EXAMINATIONS 2025

Time allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination.

Answer ***all four*** questions from this section.

Section 2 consists of three questions and carries 60% of the marks.

Answer ***two*** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

SECTION 1

Answer **all four** questions from this section.

1. Starting from the London equations,

$$\frac{\partial \mathbf{J}_s}{\partial t} = \frac{n_s e^2}{m_e} \mathbf{E},$$

$$\nabla \times \mathbf{J}_s = -\frac{n_s e^2}{m_e} \mathbf{B},$$

where $\mathbf{J}_s$ is the supercurrent density, m_e the mass of an electron and n_s the density of the electrons carrying a supercurrent, show that the ‘fluxoid’ threading a hole in a multiply-connected (torus shape) superconductor,

$$\Phi' = \int_S \mathbf{B} \cdot d\mathbf{S} + \frac{m_e}{n_s e^2} \oint_C \mathbf{J}_s \cdot d\mathbf{r},$$

is conserved, where S refers to the surface area and C refers to the contour. Under what conditions is the magnetic ‘flux’ threading the hole conserved? **[10]**

2. For a type-I superconductor, use the expression for the differential Gibbs free energy,

$$dG = -SdT - MB,$$

to derive the relationship between the superconducting condensation energy at a given temperature, $G_n(T) - G_s(T)$, and the critical field, $B_c(T)$, by making a suitable approximation. [Note that G_n and G_s refer to the Gibbs free energies in normal metallic and superconducting states, respectively.] **[10]**

3. Consider a very thin (radius = a) and long superconducting wire carrying a supercurrent. Imagine two cases where an external magnetic field is applied i) parallel to the wire and ii) perpendicular to the wire direction. For both cases, find expressions for the critical field strength H_c in terms of the applied field strength H_a and the current I that the wire carries. [Hint: you may consider demagnetisation effect using the expressions for an ellipsoid superconductor, which can be found in the Formula Sheet.] **[10]**

ANY CALCULATOR

4. Use the semiconductor model for tunneling involving superconductors (by drawing schematic diagrams) to show that the energy gap seen at zero temperature in the $I(V)$ characteristic of an SIS (Superconductor-Insulator-Superconductor) junction made from two identical superconductors is approximately twice that of the corresponding NIS (Normal metal-Insulator-Superconductor) junction. Use this model to sketch the $I(V)$ characteristic of an SIS junction made from two different superconductors at both zero temperature and finite temperature. **[10]**

SECTION 2

Answer **two** questions from this section. If you answer more than two questions, credit will only be given for the best two answers.

5. Within the framework of Ginzburg-Landau theory, the difference between the free energy per unit volume (free energy density) of the superconducting (f_s) and normal (f_n) states is given by

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla - e^*\mathbf{A})\psi|^2.$$

Minimising the total free energy leads to the Ginzburg-Landau equation

$$-\frac{1}{2m^*} (-i\hbar\nabla - e^*\mathbf{A})^2 \psi + a\psi + b|\psi|^2\psi = 0.$$

- (a) Describe the basic assumptions underlying Ginzburg-Landau theory, and explain the origin of the three terms in the formula for the free energy density. Discuss the sign and temperature dependence of the parameters a and b . **[8]**
- (b) By writing the order parameter, $\psi(\mathbf{r})$, in terms of its amplitude and phase, $\psi(\mathbf{r}) = |\psi(\mathbf{r})| \exp[i\phi(\mathbf{r})]$, show that the supercurrent may be written in terms of the superfluid velocity, $\mathbf{v}_s(\mathbf{r})$, by

$$\mathbf{J}_s = n_s^*(\mathbf{r})e^*\mathbf{v}_s,$$

where

$$\mathbf{v}_s = \frac{1}{m^*} (\hbar\nabla\phi - e^*\mathbf{A}).$$

[6]

- (c) Show that the London equations (cf. Q1) may be obtained from the equations of part (b) by making a certain approximation. Why is it possible to replace n_s^* , e^* and m^* by n_s , e and m ? **[6]**

(d) Solve the linearised Ginzburg-Landau equation,

$$\frac{1}{2m^*} (-i\hbar\nabla - e^*\mathbf{A})^2 \psi + a\psi = 0,$$

where a magnetic field is applied along z direction, with an order parameter wavefunction of the form

$$\psi(x, y, z) = \phi(x)e^{ik_y y}e^{ik_z z},$$

to find the upper critical field, B_{c2} , of a type-II superconductor in terms of the Ginzburg-Landau parameter κ . [Hint: You may use the Landau gauge $\mathbf{A} = (0, xB, 0)$.]

[10]

6. The following two equations are used to describe a Josephson junction:

$$\begin{aligned} I &= I_0 \sin \phi, \\ \frac{d\phi}{dt} &= \frac{2eV}{\hbar}. \end{aligned}$$

(a) Show that applying a DC voltage, V_0 , across a Josephson junction causes an AC current to flow across it. How is the frequency of the AC current related to the applied V_0 ? [6]

(b) Suppose that a constant current I , which is larger than I_0 , is applied to a Josephson junction connected in parallel with a resistor, R . Explain why the resulting voltage $V(t)$ will oscillate, and derive an expression for the phase $\phi(t)$. [Hint: You may find useful the indefinite integral

$$\int \frac{dx}{a - \sin x} = \frac{2}{\sqrt{a^2 - 1}} \tan^{-1} \left(\sqrt{\frac{a-1}{a+1}} \tan \frac{x}{2} \right).$$

[6]

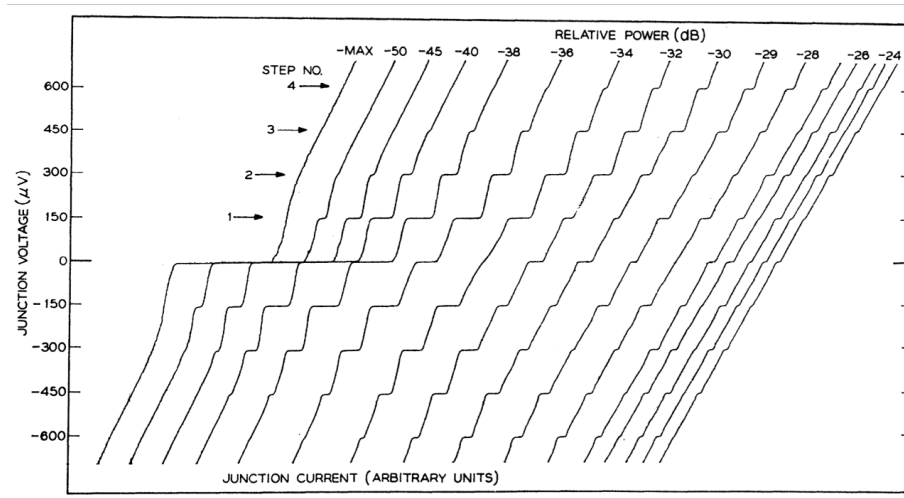
(c) Use the result of the above in (b) to find the time-averaged voltage, $\bar{V}$, across the junction, and then plot its I - V characteristic. [Hint: one does not have to explicitly evaluate $d\phi(t)/dt$ over a period, but use an expected property arising from periodicity.] [6]

(d) Suppose that a DC voltage V_0 and an AC voltage $V_1 \cos(\omega_1 t)$ are applied together across a Josephson junction. For what values of the DC voltage V_0 , will the Shapiro steps occur, and how will the half-width of the Shapiro steps vary with the AC amplitude, V_1 ? [Hint: You may find useful the formula

$$\sin(a + bx + c \sin dx) = \sum_{n=-\infty}^{\infty} (-1)^n J_n(c) \sin(a + bx - ndx),$$

where a, b, c, d are constants, and $J_n(x)$ is the n -th Bessel function.] [6]

- (e) The figure below shows the experimental V - I characteristic of a Josephson junction when DC and AC currents are applied, exhibiting the Shapiro steps. Estimate the AC frequency ω of the current source. [6]



7. The BCS ground-state wavefunction can be written as

$$|\text{BCS}\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle,$$

where $|0\rangle$ is the state vector for a vacuum and $c_{\mathbf{k}\uparrow}^{\dagger}$ is the creation operator of an electron with momentum $\hbar\mathbf{k}$ and spin up. The predictions from BCS theory include that i) the electronic specific heat C_e shows a jump at the transition temperature T_c as temperature is lowered, ii) the magnitude of the gap Δ at zero temperature is proportional to T_c , and iii) the expression for T_c is given by

$$k_B T_c = 1.14 \hbar \omega_D \exp \left(-\frac{1}{V_0 D(E_F)} \right).$$

- (a) Explain what is meant by the terms ω_D , V_0 and $D(E_F)$ in the above expression for T_c . Briefly explain the implication of this T_c expression by (i) explaining how the isotope effect manifests itself, (ii) why high conductivity metals at room temperature are often not superconducting at low temperature, while low conductivity metals become superconductors at relatively high temperature, and (iii) why $4d$ and $5d$ transition metals are often superconductors with relatively high T_c . [8]

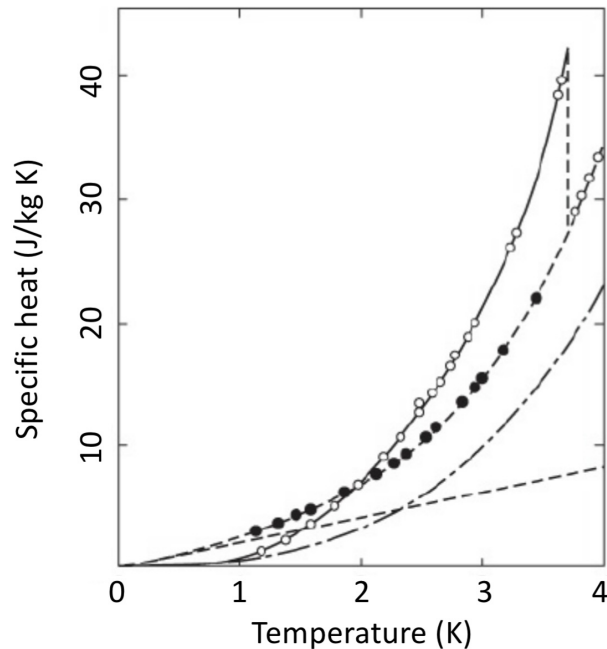
- (b) Draw curves for $|u_{\mathbf{k}}|^2$ and $|v_{\mathbf{k}}|^2$ versus momentum $\hbar\mathbf{k}$. Briefly explain the physical significance of the curves by comparing with the Fermi-Dirac distribution. [7]

- (c) Explain qualitatively the physical meaning of the following expression for the superconducting condensation energy:

$$E_s - E_n = -\frac{1}{2} D(E_F) \Delta^2,$$

where E_s and E_n are the energy in superconducting and normal metallic states, respectively. [5]

- (d) The figure below shows the specific heat of Sn measured as a function of temperature, where empty circles are the data obtained in zero field while filled circles are those in an applied field larger than the critical field [Note: Sn is a type-I superconductor].



Also plotted in the figure are the specific heat contribution from the electrons in the normal state (dashed line) and the contribution from lattice vibration (dashed-dotted line). Estimate how large the C_e jump at a superconducting transition is relative to the C_e expected for the normal state (Express as a percentage, %).

[10]

Formula Sheet

Vector identity: $\nabla \times (\nabla \times \mathbf{A}) = \nabla (\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.

Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J}, \quad (4)$$

The molar specific heat of metals at low temperature is described by

$$C = \gamma T + \left(\frac{12}{5}\right) \pi^4 R \left(\frac{T}{\Theta_D}\right)^3,$$

where γ is the Sommerfeld constant, R the gas constant, and Θ_D the Debye temperature.

For an ellipsoid superconductor, $H_i = H_a - nM$, where H_i is an internal field, H_a applied field, M magnetisation and n is a demagnetisation factor (e.g., $n = 1/3$ for a sphere, $n = 1/2$ for a long rod in a transverse field, $n = 0$ for a long rod in a longitudinal field).

Some expressions from the Ginzburg-Landau (GL) theory:

$$\lambda = \sqrt{m^* b / (4\mu_0 e^2 |a|)},$$

$$\xi = \sqrt{\hbar^2 / (2m^* |a|)},$$

$$B_{c2} = \Phi_0 / (2\pi \xi^2) = \sqrt{2} \kappa B_c,$$

where Φ_0 is the flux quantum and $\kappa \equiv \lambda / \xi$.

Some predictions from the BCS theory:

$$\xi_0 = \hbar v_F / (3.52 k_B T_c),$$

$$\Delta_0 = 1.76 k_B T_c,$$

$$k_B T_c = 1.14 k_B \Theta_D \exp[-1/D(E_F)V],$$

where $D(E_F)$ is the density of states at the Fermi level and V the effective electron-electron attraction via phonons.

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[<i>N.B.</i> 1 mole $\equiv$ 1 <i>gram-molecule</i>]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2}\text{)}$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.